FINAL INTERNSHIP REPORT

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INTERNSHIP DURATION : *2nd JUNE – 23rd JULY 2025*

TABLE OF CONTENTS

SL. NO.	CONTENT	PAGE NO.
1	ACKNOWLEDGEMENT	3
2	INTRODUCTION	4
3	INTERNSHIP ACTIVITIES	5
4	FINAL PROJECT	9
5	CHALLENGES FACED	23
6	CONCLUSION	24
7	REFERENCES	25

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to NIC iNOC, Kolkata for providing me the opportunity to undertake this internship. The experience has greatly enhanced my understanding of real-time networking systems and exposed me to the operational frameworks followed at the national level.

I am deeply thankful to **Mr. Dipankar Sarkar**, *Sr. Director (IT) and HOD – NKN and Data Center, NIC Kolkata*, for his continued support and guidance throughout my internship. I also extend my appreciation to **Mr. Manas Kumar Bhattacharya**, *Head of Department – eAbkari & Training*, for facilitating this internship and ensuring a structured learning experience.

I would like to acknowledge **Mr. Kartick Mondal**, my trainer, for his consistent mentorship and for providing relevant notes and study resources which helped me stay aligned with the objectives of the program.

I am also grateful to **Mr. Mozammil Zafar** and all the officials of NIC iNOC Kolkata and the Data Center team for their cooperation and support. Their expertise and dedication created a truly enriching and professional environment for my learning.

INTRODUCTION

The internship at the National Informatics Centre (NIC), Kolkata, was a transformative experience that offered in-depth exposure to real-world government IT infrastructure and network design. This training was instrumental in bridging the gap between academic knowledge and industry practices. The internship aimed to impart not only theoretical insights but also practical skills in network architecture, cybersecurity, protocol implementation, and system integration. Over the course of several weeks, I had the opportunity to interact with cutting-edge networking equipment, study high-level government connectivity models, and simulate enterprise-scale network topologies using Cisco Packet Tracer. This immersive experience has equipped me with a comprehensive understanding of the operational challenges and best practices involved in managing and securing large-scale networks, particularly in the context of public sector infrastructure.

INTERNSHIP ACTIVITIES

During my internship at NIC, I engaged in a diverse set of learning and practical tasks, which included:

1) *Fundamental Networking Training* :

My internship began with comprehensive training on the fundamentals of networking. I gained a strong understanding of the OSI and TCP/IP models, IP addressing (both IPv4 and IPv6), subnetting, and the functioning of essential protocols such as ARP, ICMP, DNS, DHCP, and HTTP/HTTPS. I also explored the differences between switches, routers, and hubs, and learned their configurations through hands-on practice using Cisco Packet Tracer. By performing tasks like assigning IP addresses, configuring routing tables, and simulating traffic flow, I was able to translate theoretical knowledge into practical understanding effectively.

2) *Infrastructure Exposure* :

- **NICNET**

  NICNET is the nationwide backbone network infrastructure of the National Informatics Centre. It provides seamless connectivity across ministries, departments, state governments, and district administrations. The network spans across the country through four zonal centres located in Delhi, Kolkata, Hyderabad, and Chennai. NICNET plays a crucial role in enabling e-Governance applications, secure government communication, and hosting of mission-critical infrastructure for centralized services.

- **NATIONAL KNOWLEDGE NETWORK (NKN)**

The National Knowledge Network (NKN) a high-capacity, gigabit-level network aimed at connecting India's universities, research institutions, and educational bodies. Managed by NIC under the Department of Telecommunications (DoT), it provides 1 Gbps to 10 Gbps dedicated bandwidth to over 1,600 institutes. NKN ensures high-speed collaborative research, remote computing access, and integrated use of national academic resources. It is also used to extend research capabilities to NIC's backbone network (NICNET).

- **NIC CLOUD – MEGHRAJ**

MeghRaj, also known as the NIC National Cloud, is a Government of India initiative under the National Cloud Policy. It offers scalable and secure Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS) to government departments and public service platforms. Hosted across National Data Centres in Delhi, Hyderabad, Pune, and Bhubaneswar, MeghRaj supports virtual machine hosting, containerized applications, disaster recovery, elastic resource allocation, load balancing, and CI/CD pipelines. It forms the cloud foundation for a wide range of e-Governance and citizen-centric applications.

3) ***Complex Networking Concepts Learned and Simulated Using Packet Tracer*** :

As the internship progressed, I delved into advanced networking concepts and simulated real-world topologies using Cisco Packet Tracer. I worked on configuring both static and dynamic routing protocols, specifically OSPF for intra-domain routing and eBGP for inter-domain connectivity. I practiced deploying hybrid routing architectures, combining both methods based on network requirements.

I implemented **VLANs** and **inter-VLAN routing** using Layer 3 switches, designed **DHCP** relay configurations, and ensured segmented, logically isolated sub-networks across different departments. I also configured routers for establishing WAN connectivity across simulated locations, aligning with NICNET's design. A significant part of this phase involved understanding and configuring firewalls. I simulated their real-world usage to filter traffic, configure **NAT (Network Address Translation)**, define inside and outside interfaces, and enforce traffic control using **access control lists (ACLs)**. This helped me understand how enterprise networks are secured and how devices such as firewalls, routers, and L3 switches work together to create a secure and scalable government network backbone.

I also gained an understanding of the **Spanning Tree Protocol (STP)**, a crucial loop prevention mechanism in LAN technology. STP ensures a loop-free topology by logically blocking redundant paths, which is essential for maintaining network stability and preventing broadcast storms in switched environments.

Furthermore, I explored **EtherChannel,** a link aggregation technology that bundles multiple physical Ethernet links into a single logical link. This technology is highly beneficial for increasing bandwidth, providing link redundancy, and simplifying network management, and is very useful in networking.

I learned about **Virtual Routing and Forwarding (VRF),** a networking technology that allows a single physical router to operate as multiple virtual routers, each with its own independent routing table. VRFs enhance network security and efficiency by enabling different networks to use overlapping IP addresses without conflicts. This is particularly relevant in the NIC Network, where multiple VRFs are utilized.

My learning also extended to **Multiprotocol Label Switching (MPLS)**, an efficient alternative to traditional IP routing. MPLS is especially useful for large networks, like those used by service providers and large enterprises such as the NIC Network, where it is employed to improve forwarding performance and enable various network services.

A key aspect of understanding the NIC network's role involves its connectivity to the global internet. The NICNET and National Knowledge Network (NKN) serve as critical backbones, facilitating access to the internet world for various government and academic institutions. This is typically achieved through secure gateways and robust routing configurations at the perimeter of the NIC infrastructure, allowing controlled and efficient egress to external networks while maintaining internal network integrity.

Additionally, I was introduced to the concept of adjacent **VLANs MAC binding**, a security feature that enhances network control by associating specific MAC addresses with particular VLANs on adjacent network devices. This helps in preventing unauthorized devices from connecting to sensitive network segments and maintaining strict access policies within VLANs.

4) Physical visit to NIC Data Centre

I had the opportunity to physically visit the NIC data centre, which provided me with practical exposure to live government IT infrastructure. I observed the layout of server racks, the structured cabling system, high-capacity UPS units, and redundant cooling systems. I witnessed how enterprise-grade network devices such as **routers**, **core and distribution switches**, **firewalls**, and **load balancers** are physically deployed and managed. I also gained insights into the command-and-control systems used to monitor these setups in real-time. This experience helped me understand the physical and operational complexities behind maintaining high-availability government networks.

FINAL PROJECT

Simulated National Informatics Centre (NIC) Network Architecture

Introduction:

During my internship at the National Informatics Centre (NIC), Kolkata, in the domain of Computer Networking and Communication, I undertook a capstone project aimed at simulating a comprehensive government-level networking infrastructure using Cisco Packet Tracer. This project provided an invaluable opportunity to apply theoretical knowledge to a practical, real-world scenario, focusing on the design and implementation of a robust and scalable network connecting geographically dispersed offices. The insights gained from this simulation are crucial for understanding the complexities and operational demands of large-scale governmental networks.

Project Objective

The primary objective of this project was to design, implement, and simulate a robust, scalable, and resilient network infrastructure mirroring the National Informatics Centre (NIC)'s operational requirements, utilizing Cisco Packet Tracer. This simulation aimed to demonstrate advanced networking concepts, including inter-Super PoP connectivity, regional network integration with redundant paths, and the implementation of various routing protocols and services to ensure seamless, efficient, and highly available communication across a geographically distributed governmental network.

Network Overview and Topology

The simulated network architecture represents a hierarchical design, connecting the four major NIC Super PoPs located in Kolkata, Delhi, Chennai, and Hyderabad. Each Super PoP acts as a central hub, extending connectivity to its respective regional networks (NICNET).

The core of the network features a full-mesh topology connecting the four Super PoPs, ensuring robust inter-city redundancy. From each Super PoP, connections extend to multiple regional routers. While the simulation primarily focused on four, **it's important to note that the NIC network features six Super PoPs across the country**, providing extensive connectivity. A key design element in the regional segments is the **triangular network connection formed between any two regional routers and their respective Super PoPs.** This configuration creates a redundant path, ensuring that if one link or regional router connection fails, both regional routers and their connected LANs can still be accessed from the Super PoPs. This significantly enhances network resilience and availability. End-user devices (PCs) are then connected to these regional routers, often through switches, simulating departmental or office environments.

Key Components and Technologies Implemented

This project leveraged a variety of Cisco networking devices and protocols to build a comprehensive and functional simulated environment:

- ***Routers:*** Utilized as the backbone for inter-Super PoP and Super PoP-to-regional connectivity.

- ***Layer 3 Switches:*** Employed in specific regional networks (e.g., under Kolkata Super PoP) to facilitate VLAN segmentation and inter-VLAN routing.
- ***Layer 2 Switches:*** Used to connect end-user devices within local area networks.
- ***End Devices:*** Personal Computers (PCs) simulating user workstations.

The following networking technologies and protocols were meticulously configured:

- ***External Border Gateway Protocol (eBGP):***
    - Configured among the four NIC Super PoPs (Kolkata, Delhi, Chennai, Hyderabad) to establish robust and scalable inter-Super PoP routing. eBGP ensures efficient and policy-based routing of traffic between these major central points of the NIC network.
- ***Open Shortest Path First (OSPF):***
    - Implemented to manage routing within the regional networks (NICNET) connected to each Super PoP. OSPF was chosen for its rapid convergence and efficient routing in internal network segments, particularly benefiting from the redundant paths within the regional triangular topology.
    - ***Redistribution:*** To enable end-to-end communication across the entire simulated network, routes learned via

OSPF within regional networks were redistributed into eBGP at the Super PoP routers, and conversely, eBGP routes were redistributed into OSPF for regional access to the core network. This critical configuration ensures seamless reachability between any two points in the simulated NIC network.

- ***Static IP Addressing:***
  - The majority of the network, including inter-Super PoP links and most regional networks (Kolkata, Delhi, and Chennai), utilizes static IP addressing. This approach provides predictable addressing and simplified management for critical backbone and stable regional segments.
- ***Dynamic Host Configuration Protocol (DHCP):***
  - The regional routers under the Hyderabad Super PoP were specifically configured to act as DHCP servers. This demonstrates the dynamic allocation of IP addresses to end devices within these regional networks, showcasing a common practice for flexible and scalable user access.
- ***Static Routing:***
  - A specific design choice was made for the Chennai regional network (LAN). Since there is only a single logical path between the main Chennai Super PoP router and its immediate regional routers, OSPF was intentionally omitted for this segment. Instead, static routes were configured, demonstrating an efficient and resource-light routing solution for simple, point-to-point connections where dynamic routing overhead is unnecessary.

- ***Virtual Local Area Networks (VLANs) & Inter-VLAN Routing:***
    - In two of the regional routers under the Kolkata Super PoP, Layer 3 switches were deployed. These switches were configured with multiple VLANs to segment the local network into different logical broadcast domains, enhancing security and broadcast control. Inter-VLAN routing was then configured on these Layer 3 switches to enable communication between devices residing in different VLANs within the same regional segment.

Functionality and Demonstrations

This comprehensive simulation successfully demonstrated several critical functionalities vital for a government-level networking infrastructure:

- ***Inter-Super PoP Connectivity:*** Seamless communication between all four NIC Super PoPs using eBGP, proving a resilient and scalable backbone.
- ***Super PoP-to-Regional Connectivity with Redundancy*:** Effective routing between Super PoPs and their associated regional NICNETs via OSPF, critically showcasing the ***redundant triangular path design.*** This ensures that if any single link connecting a regional router to the Super PoPs fails, communication to both regional routers in that cluster remains uninterrupted through the alternative path.
- ***Full Network Reachability:*** Due to careful redistribution between OSPF and eBGP, all end devices across the entire

simulated NIC network can communicate with each other, regardless of their physical location or the routing protocol used in their segment.

- ***Dynamic IP Assignment:*** Demonstrated the practical application of DHCP for automated IP address management in specific regional networks (Hyderabad).
- ***Efficient Static Routing*:** Illustrated the utility of static routing for simplified topologies (Chennai LAN).
- ***Network Segmentation and Security (VLANs):*** Showcased the use of VLANs for logical separation and enhanced security within regional LANs, with Layer 3 switches facilitating necessary inter-VLAN communication.

Challenges Faced and Solutions Implemented

Implementing this complex network simulation presented several key challenges, each contributing to a deeper understanding of network design and troubleshooting:

- ***eBGP and OSPF Redistribution:*** The primary challenge was ensuring seamless route exchange and preventing loops when redistributing routes between eBGP (inter-Super PoP) and OSPF (regional). This was overcome by carefully configuring redistribution commands on border routers, paying close attention to administrative distances and metrics, and iteratively verifying routing tables for optimal path selection.

- ***Redundant Path Convergence:*** Designing redundant triangular paths within regional networks required ensuring OSPF's efficient convergence for failover. This was addressed

  by leveraging OSPF's fast convergence properties and verifying successful traffic rerouting through alternative paths during simulated link failures.

- ***Inter-VLAN Routing on Layer 3 Switches:*** Correctly configuring VLANs and inter-VLAN routing on Layer 3 switches in the Kolkata region demanded meticulous setup of SVIs and trunk links. Troubleshooting involved detailed verification of VLAN assignments and routing between logical segments.

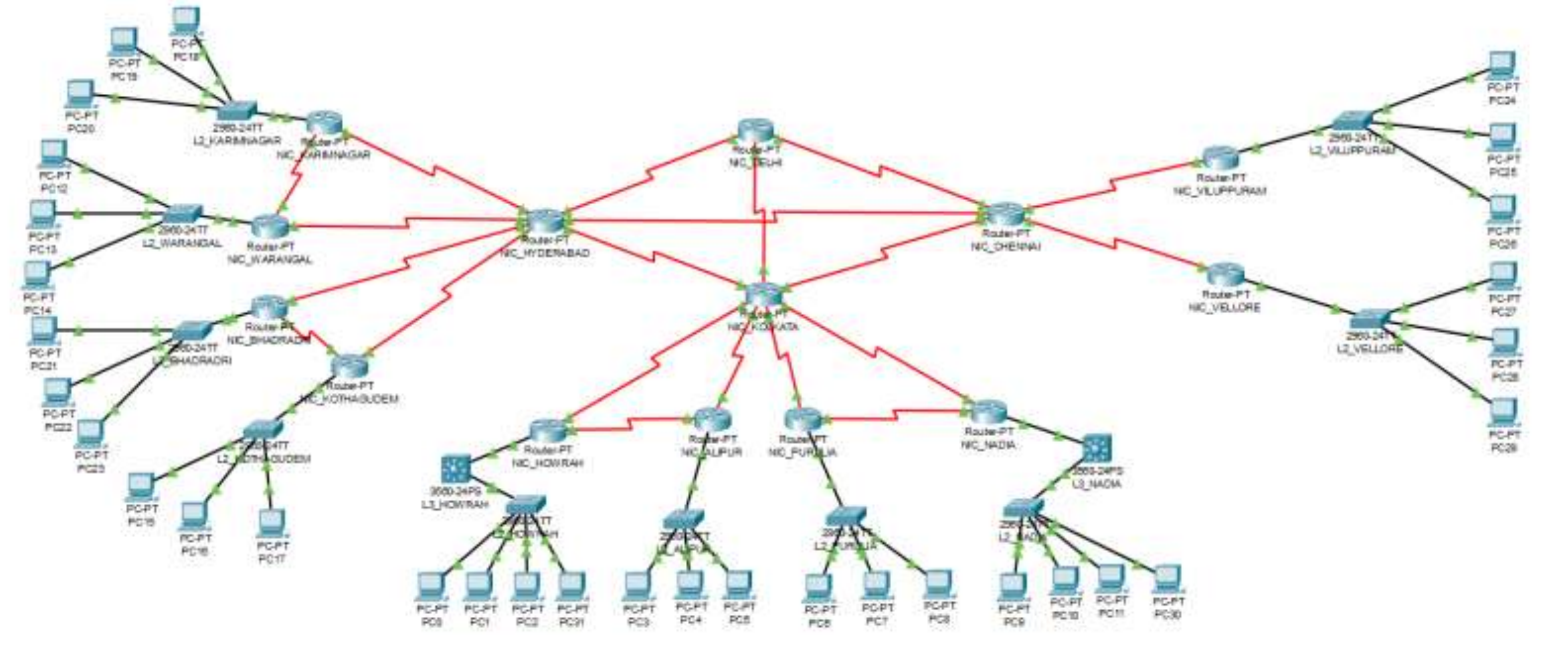
- ***IP Addressing Management:*** Managing a large and mixed IP addressing scheme (static and DHCP) across the entire network was a consistent challenge. This was resolved by creating a comprehensive pre-planned addressing scheme and carefully configuring DHCP pools to prevent overlaps and ensure efficient resource allocation.

Overcoming these challenges significantly enhanced my practical troubleshooting and problem-solving skills, crucial for real-world network deployments.

Visual Evidence: Network Topology and Verification :

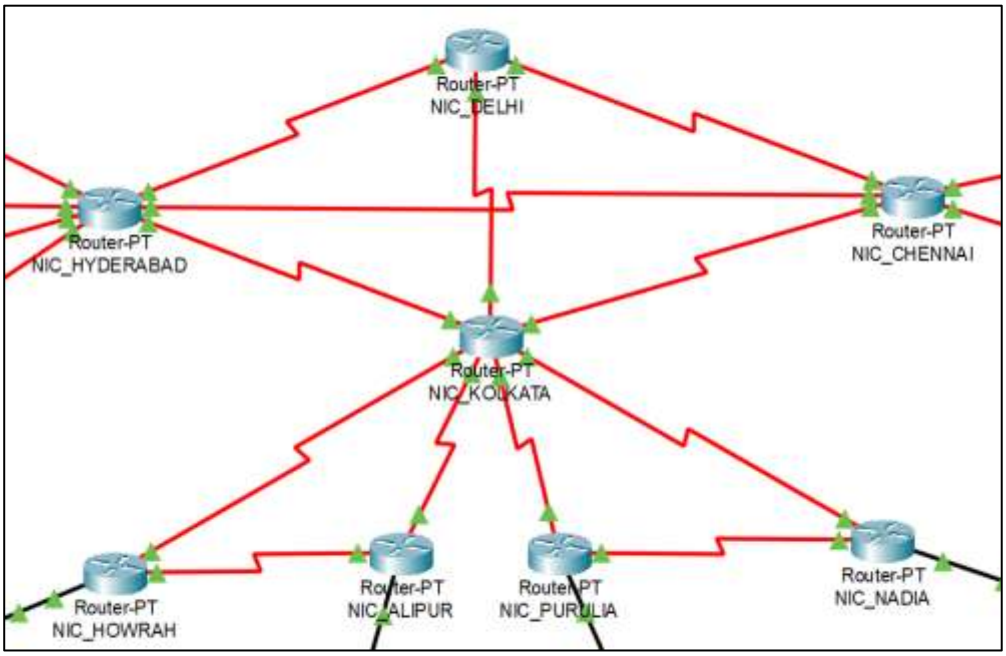
This section provides visual documentation of the simulated NIC network, including the overall topology and verification of network connectivity and configuration through command-line outputs from end devices.

1) *Overall Network Topology:*

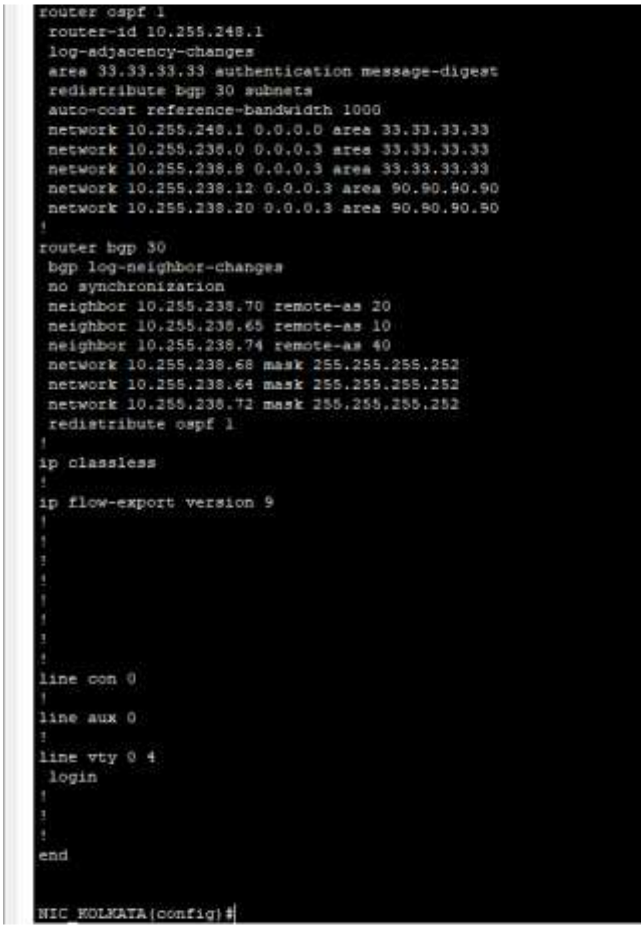


> ➢ *Overview of the simulated National Informatics Centre (NIC) network architecture. This comprehensive topology illustrates the core eBGP connections between Super PoPs and the OSPF-enabled regional networks, including the critical triangular redundant paths designed for high availability.*

2) *Detailed Super PoPs Connectivity (Example: Kolkata Super PoP):*



```
!
interface Loopback1
 ip address 10.255.248.1 255.255.255.255
!
interface Serial0/0
 description connected to NIC_HOWRAH
 ip address 10.255.238.1 255.255.255.252
 ip ospf message-digest-key 1 md5 NIC
 ip ospf network point-to-point
 clock rate 2000000
!
interface Serial1/0
 description connected to NIC_ALIPUR
 ip address 10.255.238.9 255.255.255.252
 ip ospf message-digest-key 1 md5 NIC
 ip ospf network point-to-point
 clock rate 2000000
!
interface Serial2/0
 description connected to NIELIT
 ip address 10.255.238.13 255.255.255.252
 ip ospf message-digest-key 1 md5 NIC
 ip ospf network point-to-point
 clock rate 2000000
!
interface Serial3/0
 description connected to JU
 ip address 10.255.238.21 255.255.255.252
 ip ospf message-digest-key 1 md5 NIC
 ip ospf network point-to-point
 clock rate 2000000
!
interface Serial4/0
 description connected to NIC_DELHI
 ip address 10.255.238.66 255.255.255.252
!
interface Serial5/0
 description connected to NIC_HYDERABAD
 ip address 10.255.238.69 255.255.255.252
 clock rate 2000000
!
interface Serial6/0
 description connected to NIC_CHENNAI
 ip address 10.255.238.73 255.255.255.252
!
```

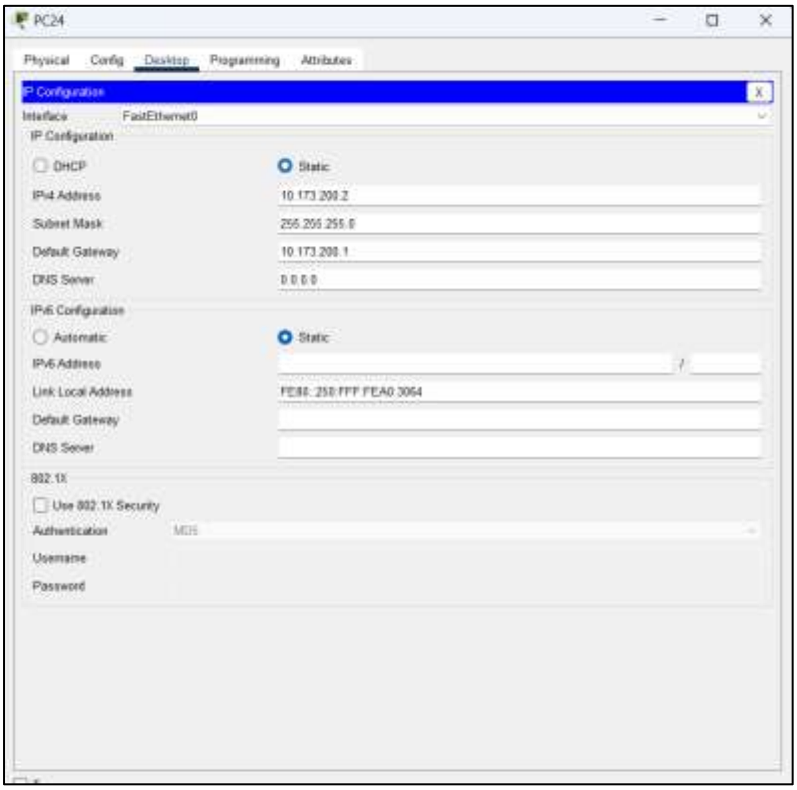


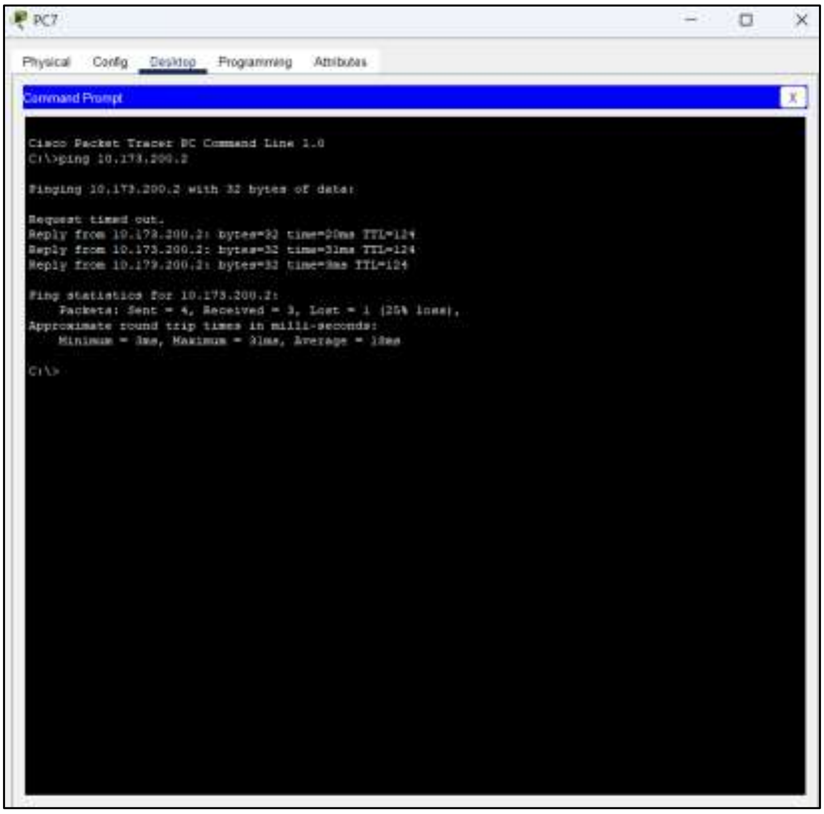
```
router ospf 1
 router-id 10.255.248.1
 log-adjacency-changes
 area 33.33.33.33 authentication message-digest
 redistribute bgp 30 subnets
 auto-cost reference-bandwidth 1000
 network 10.255.248.1 0.0.0.0 area 33.33.33.33
 network 10.255.238.0 0.0.0.3 area 33.33.33.33
 network 10.255.238.8 0.0.0.3 area 33.33.33.33
 network 10.255.238.12 0.0.0.3 area 90.90.90.90
 network 10.255.238.20 0.0.0.3 area 90.90.90.90
!
router bgp 30
 bgp log-neighbor-changes
 no synchronization
 neighbor 10.255.238.70 remote-as 20
 neighbor 10.255.238.65 remote-as 10
 neighbor 10.255.238.74 remote-as 40
 network 10.255.238.68 mask 255.255.255.252
 network 10.255.238.64 mask 255.255.255.252
 network 10.255.238.72 mask 255.255.255.252
 redistribute ospf 1
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 login
!
!
!
end

NIC_KOLKATA(config)#
```

- *Detailed view of the Kolkata Super PoP network segment. This highlights the connections to its regional routers and the implemented triangular topology, clearly showing the redundant pathways designed to maintain connectivity even in case of a single link failure. The specific IP addresses and port configurations are visible, alongside indicators of active OSPF and BGP processes on the router.*

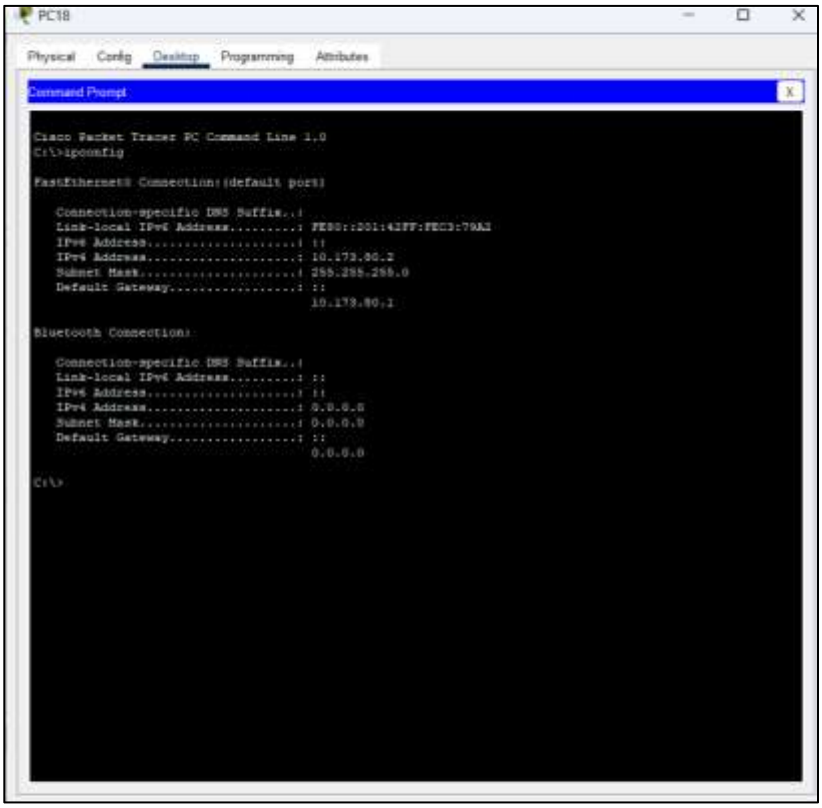
3) <u>Command Prompt Verification (Ping Test - Inter-Super PoP/Regional)</u> :





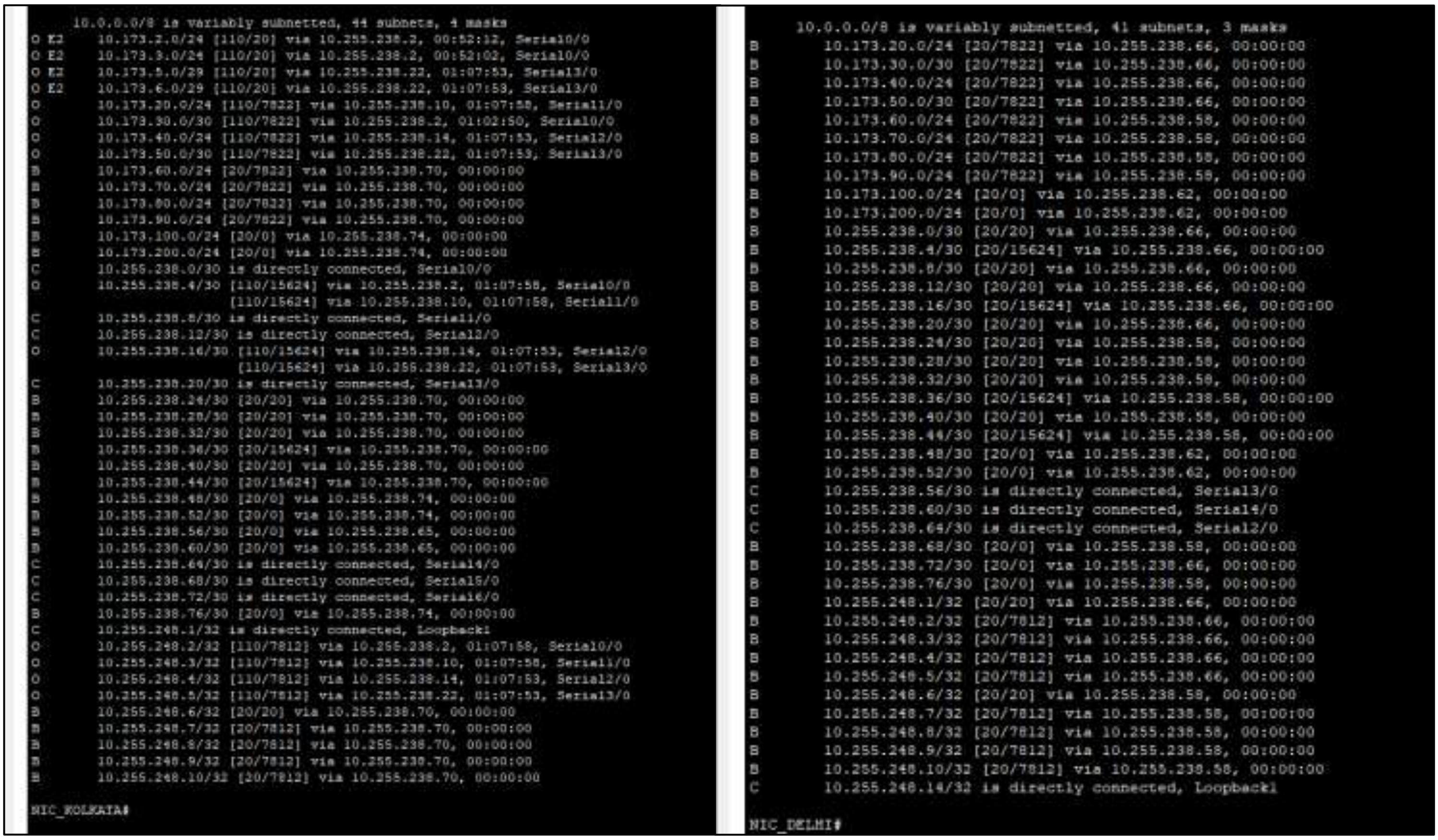
- *Verification of end-to-end connectivity across the network. A successful ping from a PC in the Kolkata regional network to a PC in the Chennai regional network demonstrates seamless routing across the eBGP and OSPF domains, confirming full network reachability.*

4) Command Prompt Verification (IP Configuration - DHCP Example) :



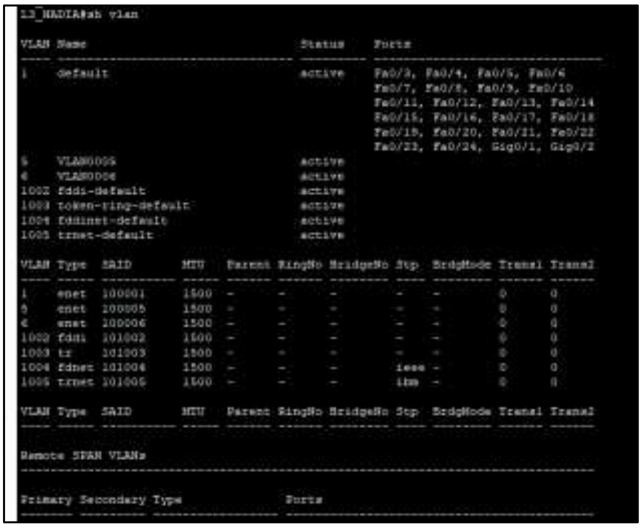
- *Dynamic IP address assignment verification. The IP configuration details from a PC in the Hyderabad regional network confirm that the device has successfully obtained its IP address, subnet mask, and default gateway via DHCP from the regional router.*

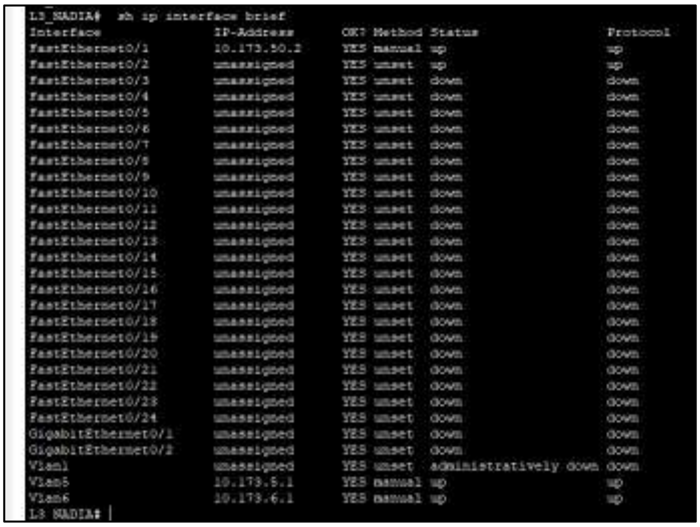
5) Router Routing Table (Example: Delhi and Kolkata Super PoP Router) :

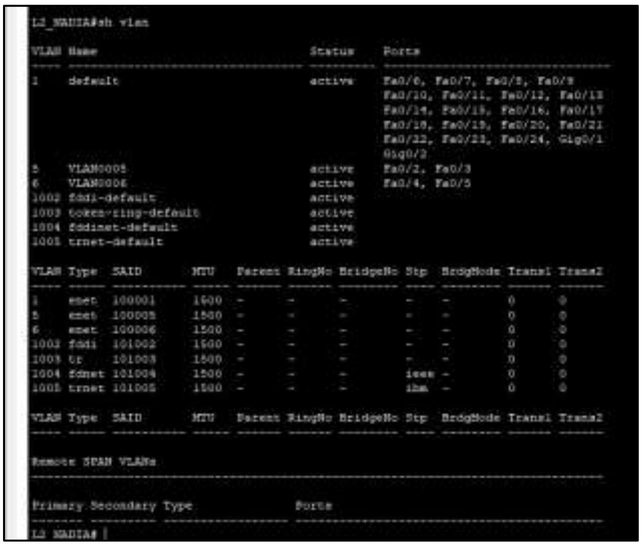


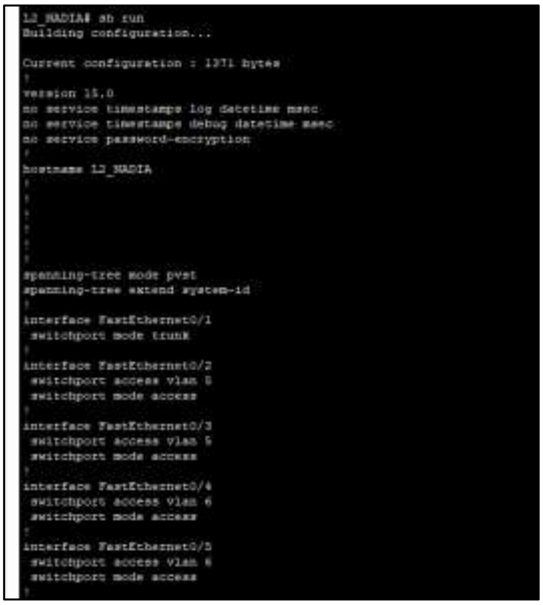
- *Routing table of the Delhi and Kolkata Super PoPs router. This output displays routes learned from directly connected networks (denoted by 'C'), eBGP (denoted by 'B') for inter-Super PoP connectivity, and OSPF (denoted by 'O') for its regional networks. The presence of routes from all relevant sources confirms successful configuration and redistribution, enabling comprehensive network communication and demonstrating redundant path awareness.*

6) <u>VLAN Configuration Verification (Nadia Layer 3 Switch)</u> :









- *VLAN implementation details within the Kolkata regional network. The outputs from the Layer 3 switch verify VLAN configurations and their corresponding Switch Virtual Interfaces (SVIs) for inter-VLAN routing, while the Layer 2 switch outputs (demonstrate individual port assignments to access VLANs and the configuration of trunk ports facilitating multi-VLAN traffic flow to the Layer 3 switch.*

Learnings and Outcomes :

This final project significantly enhanced my understanding and practical skills in complex network engineering, crucial for large-scale government networks. Key outcomes include:

- ***Advanced Routing Protocols*:** Gained proficiency in configuring and troubleshooting eBGP and OSPF, including route redistribution.
- ***Network Design:*** Applied hierarchical and redundant network design principles for scalability and fault tolerance.
- ***IP Addressing:*** Mastered both static and dynamic IP addressing strategies.
- ***LAN Segmentation*:** Acquired expertise in VLANs and Layer 3 switching for network segmentation and inter-VLAN routing.
- ***Troubleshooting:*** Developed strong systematic troubleshooting skills in a multi-protocol environment.
- ***Cisco Packet Tracer:*** Deepened proficiency in using Packet Tracer for network simulation and validation.

This project provided invaluable hands-on experience, solidifying my understanding of real-world networking challenges and solutions within a government organizational context.

CHALLENGES FACED

During my internship at NIC, I faced several technical challenges that enhanced my understanding of networking concepts. Initially, configuring OSPF and eBGP in Cisco Packet Tracer was difficult, especially in ensuring proper route exchange and stable connectivity across regions.

Managing VLANs and Layer 3 switch configurations also required careful planning to enable inter-VLAN communication and trunking. Debugging connectivity issues due to IP mismatches or missing routes helped me build a systematic troubleshooting approach.

While working on the final simulation project, integrating static and dynamic routing protocols under a single topology was complex. Assigning IPs without conflicts and ensuring seamless connectivity between regions and DHCP-based end devices demanded detailed attention.

Additionally, the limitations of Packet Tracer sometimes restricted real-world configurations, requiring smart adaptations. These challenges greatly contributed to the development of my practical skills and confidence in handling network simulations.

CONCLUSION

My internship at the National Informatics Centre (NIC), Kolkata, has been an immensely valuable and career-shaping experience. It allowed me to go beyond theoretical knowledge and engage directly with real-world networking infrastructures and operations.

Throughout the internship, I developed a strong understanding of core protocols such as OSPF, eBGP, static routing, and DHCP, along with practical skills in VLAN creation and Layer 3 switching. I also explored advanced concepts like MPLS, VTP, STP, DTP, and VRF. While not all were implemented in the final project, gaining insight into their role in large-scale networks broadened my perspective.

Exposure to infrastructures like NICNET, NKN, and MeghRaj helped me appreciate the scale and importance of secure government networks. The physical visits to the data center and observation of key networking devices further enhanced my learning.

The final project—simulating NICNET backbone with dynamic and static routing, VLAN segmentation, and DHCP—challenged me to apply my skills effectively and strengthened my ability to design, configure, and troubleshoot networks.

This internship has greatly enhanced my technical capabilities and provided clarity on my future direction in the fields of communication systems and network design.

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- **Networking Concepts – Complete Course** (YouTube)
  Video: "Computer Networking Full Course" by Simplilearn
  https://youtu.be/IPvYjXCsTg8

- **Government of India NKN Overview** – National Knowledge Network
  https://nkn.gov.in

- Study notes, handouts, and training presentations provided by **Mr. Kartick Sir** during NIC internship sessions.